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Phase-field modelling and simulation of morphology transition induced by thermoelectric effects during solidification in pure metals

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Abstract: Appling direct current (DC) on solidifications has become an efficient way to modify the solidification process by altering the heat and solute diffusions during phase transitions. In this study, we present a phase-field model for simulating the microstructure evolution in pure metals with applied DC. Effect of different magnitude DC on the planar evolution as well as dendrite growth in pure Sn and Bi is simulated. Results indicate that DC has very different influences on solid/liquid interface migrations. These morphology transitions can be directly attributable to temperature gradient changes in the growing direction. Thus, the thermoelectric heat (including the Joule heat, Thomson heat and Peltier heat) induced by applied DC as well as the latent heat release during liquid-solid transition is calculated, and their effects on the temperature distributions near solidifying interfaces is discussed. Therefore, applied DC has a signification effect on the microstructure formations in pure metals by changing the stability of the solid/liquid interfaces with thermoelectric effect.

Keywords: Phase field method; Thermoelectric effects; Interface stability; Microstructure

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1. Introduction

Dendritic growth is a fundamental crystal growth phenomenon accompanying most casting and solidification processes, where it is detrimental to crystalline quality[1-4], thus understanding dendritic growth is considered essential for controlling basic solidification. The subject of dendritic growth will be presented on the basis of heat and mass transfer, capillarity effects at the solid/melt interface, and interfacial dynamics, including morphological stability, and side-branching dynamics. In pure metals, according to the principle of solidification, the only source of growing interface instabilities is the thermal undercooling. When the thermal gradient in the liquid at the solid/liquid interface is less than the liquidus temperature gradient, this liquid is constitutionally undercooled. Instabilities growing in this region will become stable, since they will continue to grow into the undercooled liquid. On the contrary, the interface will remain planar. Thus, it is necessary to quantify the shape and position of the solid/liquid interface versus time to have a good knowledge of solidification.

In the last decades, the application of direct current (DC) has been widely used to control and modify the solidification process[5-7]. Due to thermal phenomena including Joule heating, Peltier and Thomson effects, etc. heat may be emitted or absorbed during crystal solidification with an electric current passing through the sample. Theoretical studies have conducted by calculating the effect of the current on morphological instability of solidifying interfaces to study the motion of a crystal/melt interface, considering the Thomson effect, Peltier effect and Joule heating[8-13]. Most recently, the effect of DC on the microstructures of pure metals as well as alloys has been widely investigated, some inconsistent experimental phenomena have been achieved[14-17]. Furthermore, the potential mechanism of DC on the

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microstructure modification is still controversial because of the complicated effects induced by the electric field.

Nowadays, phase-field modelling and simulation methods have become an efficient way to study the microstructure evolution as well as a variety of physical effects accompanied with phase transitions[18-20]. Therefore, in this study, a phase-field model for solidification in pure metals with thermoelectric effect (including Joule heat effect, Thomson effect and Peltier effect) is presented based on the physical models for thermoelectric effects[8, 12]. The effect of current-induced thermoelectric processes on the evolution of growing solidification patterns is simulated and discussed. It is believed that the solid/liquid interface migration is highly dependent on the temperature distributions along solidifying directions. Thus the Joule heat, Thomson heat, Peltier heat as well as latent heat release are calculated and analyzed to examine their effects on the interface stability during solidification.

2. Phase-field model

Based on Kobayashi's model for dendrite growth in pure melt[21], we involve the thermoelectric effects into the heat diffusion equation to study the dendrite growth phenomena in undercooled melt under direct current. In two dimensional study, the governing equation for the phase-field variable $\phi(\boldsymbol{r}, t)$ (in solid phase ϕ=1, in liquid phase ϕ=0) is

$$\tau_\phi \frac{\partial \phi}{\partial t} = \phi(1-\phi)\left(\phi - \frac{1}{2} + m + a\chi\right) + \nabla\cdot\left(\varepsilon^2 \nabla\phi\right) - \frac{\partial}{\partial x}\left(\varepsilon\varepsilon' \frac{\partial \phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\varepsilon\varepsilon' \frac{\partial \phi}{\partial y}\right) \quad (1)$$

where τ_ϕ is a small positive constant, $m(T) = \frac{\alpha}{\pi}\tan^{-1}\left[\gamma\left(T_e - T\right)\right]$ where T_e is an equilibrium temperature and α is a positive constant satisfying α<1, since this assures $\left|m(T)\right| < 1/2$ for all values of temperature T. a is the amplitude of the noise and χ is a random number sequence uniformly distributed on the interval [-0.5, 0.5]. ε represents the

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gradient energy induced by the solid-liquid interface, and $\varepsilon'=\partial\varepsilon/\partial\theta$, when we take $\varepsilon=\varepsilon(\theta)$ where θ is the angle between the interface normal and the orientation of the crystal lattice which can be calculated by $\beta=\arctan\left(\left(\partial\phi/\partial y\right)/\left(\partial\phi/\partial x\right)\right)$.

In accordance with discussions in[8, 12], the energy equation for heat diffusion involving thermoelectric effects can be written as

$$\frac{\partial}{\partial t}\left(\rho CT\right)=\nabla\cdot\left(\kappa\nabla T\right)+\frac{J^2}{\sigma}-\tau\boldsymbol{J}\cdot\nabla T+\pi\boldsymbol{J}+L\rho\frac{\partial\phi}{\partial t} \tag{2}$$

where ρ is the density, C is the heat capacity, κ is the heat conductivity, $\boldsymbol{J}$ represents the current density, σ is the electric conductivity, τ is the Thomson coefficient, π is the Peltier coefficient ($\pi=(s_L-s_S)T_M$; s_L and s_S are Seebeck coefficients for liquid and solid phase, respectively, and T_M is the melting temperature), and L is the latent heat of liquid-solid phase transition. Hence, terms on the right side of equation (2) represent the heat conduction, Joule heat, Thomson effect, Peltier effect and latent heat, respectively.

For liquid-solid phase transition, the physical properties in for solid and liquid are different, thus Equation (2) should be modified to involve the coefficient differences between the solid and liquid. In this study, we assume a simply linear variation for these parameters:

$$C=C_L+\phi\left(C_S-C_L\right) \tag{3}$$

$$\kappa=\kappa_L+\left(\kappa_S-\kappa_L\right)\phi \tag{4}$$

$$\sigma=\sigma_L+\left(\sigma_S-\sigma_L\right)\phi \tag{5}$$

$$\tau=\tau_L+\left(\tau_S-\tau_L\right)\phi \tag{6}$$

The corner marks S and L represent liquid and solid phase, respectively. All these assumptions assure that C, κ, σ, τ take the value C_S, κ_S, σ_S, τ_S in solid(ϕ=1) and C_L, κ_L, σ_L, τ_L in liquid(ϕ=0). The density difference between the solid and liquid is not considered in this

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model($\rho=\rho_S=\rho_L$) , and the convection is not involved in this model either.

Equations (1) and (2) are nondimensionalized and solved via the Finite Difference Method. Zero-flux boundary conditions are applied, and the model parameters for the simulation are present in Table 1.

Table 1 Parameters used in simulations

Parameter	ε_0	τ	α	γ	a	δ	j	$\Delta x=\Delta y$	Δt	K
Value	0.01	0.0003	0.9	10.0	0.005	0.04	4	0.03	0.0002	1.5

From equation (2), we can find that the effect of the current on the heat diffusion during solidification is highly correlated to the current direction as well as the physical property differences between the solid and liquid phase. So in this study, we take two different pure metals, Sn and Bi, to investigate effect of direct current on the morphology evolution. Parameter values for the simulations are listed in Table 2.

Table 2 Physical parameters for Sn and Bi[8].

	Sn	Bi
Melting temperature T_M (K)	505	504
Latent heat L (J/m^3)	4.3×10^8	5.1×10^8
Heat capacity liquid C_L (J/m^3/K)	1.71×10^6	1.42×10^6
Heat capacity solid C_S (J/m^3/K)	1.95×10^6	1.34×10^6
Thermal conductivity liquid κ_L (W/m/K)	30.3	14.31
Thermal conductivity solid κ_S (W/m/K)	59.5	7.05
Electric conductivity liquid σ_L (S/m)	4.35×10^6	3.85×10^5
Electric conductivity solid σ_S (S/m)	2.08×10^6	7.81×10^5
Thomson coefficient liquid τ_L (V/K^2)	-9×10^{-9}	-1×10^{-8}
Thomson coefficient solid τ_S (V/K^2)	2.7×10^{-9}	8×10^{-8}
Seebeck coefficient liquid s_L (V/K)	-0.56×10^{-6}	0
Seebeck coefficient solid s_S (V/K)	-1.69×10^{-6}	-5.3×10^{-5}

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3. Results and discussions

3.1 Planar evolution with electric current

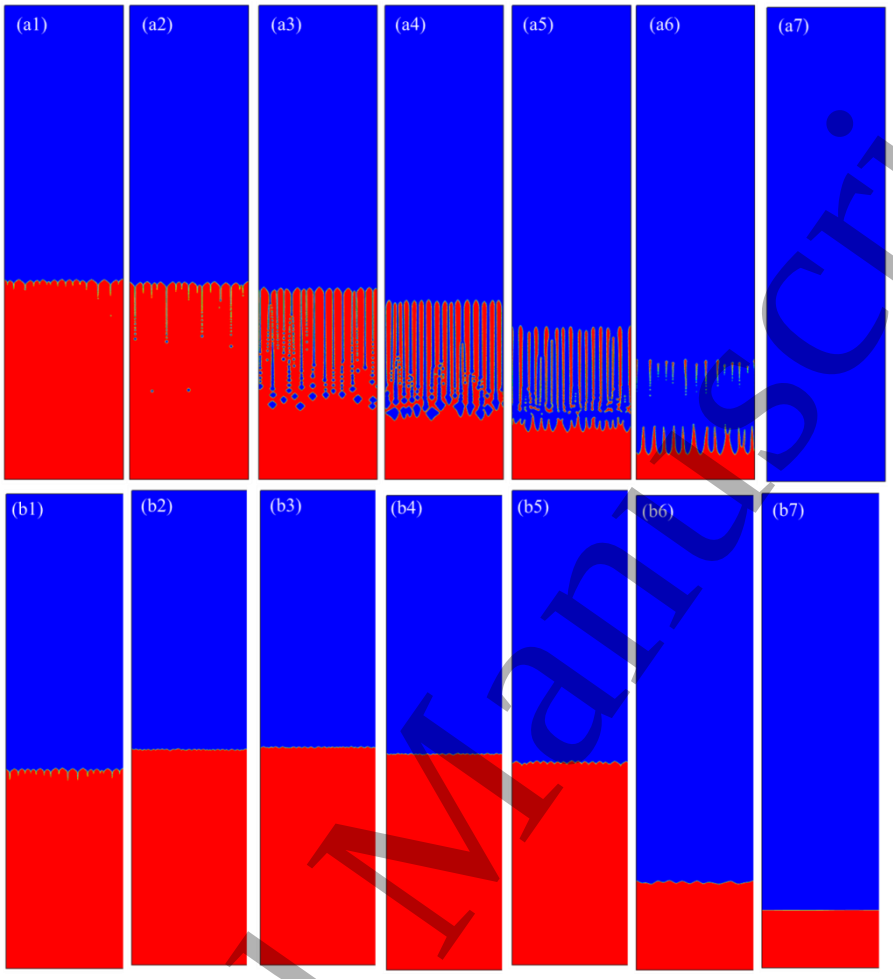


Figure 1 Morphology at the same time step induced by different electric current density. a1-a7 is the morphology formed in Sn melt with current density of 0, 5 A/m^2, 10 A/m^2, 15 A/m^2, 20 A/m^2, 25 A/m^2, 30 A/m^2, respectively. b1-b7 is the morphology formed in Bi melt with current density of 0, 5 A/m^2, 10 A/m^2, 15 A/m^2, 20 A/m^2, 25 A/m^2, 30 A/m^2, respectively.

In this section, we focus on the heat effect induced by DC on the planar solid/liquid interface migration during solidifying in pure metals, and the morphology of the solid phase formed at the same time step with different current densities is given in Fig.1. It is obvious that for Sn and Bi, the morphology changes show different characteristics, which should be

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correlated to the thermal effect induced by the current. For solidification without current, weak cellular structure is formed as discussed in Kobayashi's report[21]. For pure Sn solidifying, with current density increasing, tip splitting occurs and the splitting branches tend to spread widely with the trend to form a columnar structure. Columnar arms keeps shrinking with increasing current density and at a certain point, and remelting broken occurs with extremely large current density. The whole region would be remelted and the solid phase would vanish. For pure Bi solidifying, the morphology change induced by increasing current dentistry shows a different feature: with increasing current density, the solid-liquid interface migration is slightly enhanced firstly and then restrained with large current densities.

The planar interface growth rate is largely changed with induced current, showing different characteristics for Sn and Bi. The solid-liquid interface positions as a function of current densities plotted in Fig.2, and it can be obviously found that the current direction has a relatively small effect on the solidifying rate, which is to say, the Thomson effect shows a slight influence on the temperature distributions, and Joule heat and Peltier might be main factors leading to the interface migration changing, which determines the solid/liquid interface morphology during solidifying with applied DC.

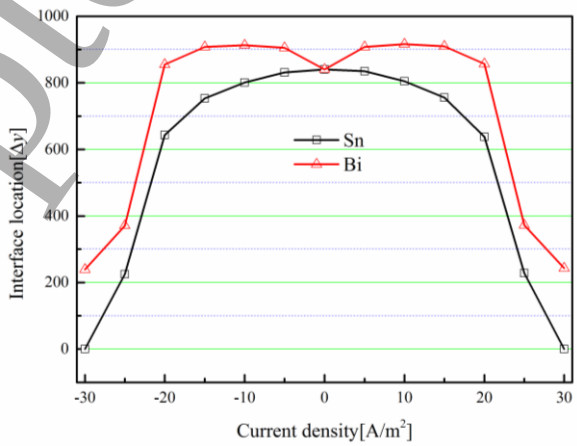


Figure 2 Solid-liquid interface position with different electric current density.

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As stated in fundamentals of solidifications, the phase stability during solidifying in a pure substance is highly correlated with the temperature gradient at the solid-liquid interfaces, thus the growth rate is controlled by the diffusion of latent heat away from the advancing crystal–melt interface. In Fig.3, the temperature distribution profiles along the solidifying direction is plotted for different current densities according to the simulation in Fig.1. The temperature gradient $G = dT / dy$ is varying with current densities as well as metal properties, and the temperature gradient for each case is consistent with the interfaces morphology. Columnar grains grow out from the mound (which is the heat sink) in a direction which is opposite to that of the heat flow, which has been generally validated in experimental researches and theoretical studies. The solid-liquid interface of a pure metal will always be stable if the temperature gradient is positive, and unstable if the gradient is negative[22].

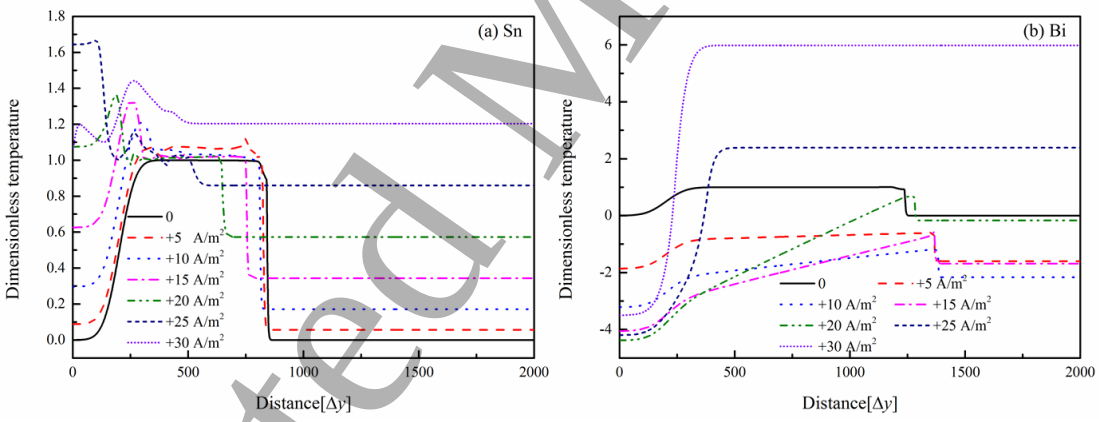


Figure 3 Temperature profiles along the solidifying direction.

It is well established that, for pure metal solidifying, the interface stability depends on the direction of heat flow. Thus, the interface stability during solidifying is mostly determined by the temperature distributions near the interfaces[12]. When a perturbation forms at an initially smooth interface, the temperature gradient in the liquid increases while the gradient in the solid in proportional to the gradient, leading more heat flowing into the solid. As a

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result, the perturbation melts back and the planar interface is stabilized. Therefore, the planar structure formed with applied DC is determined by the temperature gradient near the solid/liquid interface, which is highly dependent on the DC magnitude and metal physical parameters, as shown in Fig.1.

3.2 Dendrite growth with electric current

Dendritic growth is another fundamental crystal growth phenomenon in solidification, which requires the coupling of two independent growth processes: (1) the steady-state evolution of the dendrite tip, and (2) the non-steady-state development of dendrite branch[1]. One of the most interesting phenomena associated with thermoelectric effects at the interface is that the internal electric current due to the Seebeck effect always causes Peltier cooling at the tip of a dendrites, which is a destabilizing effect[23]. As a result, the Seebeck and Peltier effects together play a role on the interracial stability which is systematically destabilizing.

In this section, the dendritic structure formation with applied DC is simulated, and the morphologies at the same time step with different current densities is plotted in Fig.4. It can be easily seen that applied DC has obviously different influences on the dendrite morphologies for Bi and Sn. With the DC increasing, the Sn dendrite shows a strong side-branching transition: the side-branches are firstly enhanced with small DC, then remelted with increasing DC, and melted away with large DC. For Bi, the equiaxed grain is firstly changed to globular with small DC, then changed back to equiaxed, and finally melted away with large DC. No side-branching is detected during solidification of Bi equiaxed grains. The growth rate of the equiaxed grains is also greatly changed with increasing DC, and no solid would be found with large DC due to the large Joule heat effect. Secondary and tertiary

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branches are important insofar as they establish the length scales and pattern which is of immense practical importance in determining the engineering properties of materials that solidify dendritically. Thus applying DC into solidification might be an efficient way to adjust the dendritic morphology in pure metals.

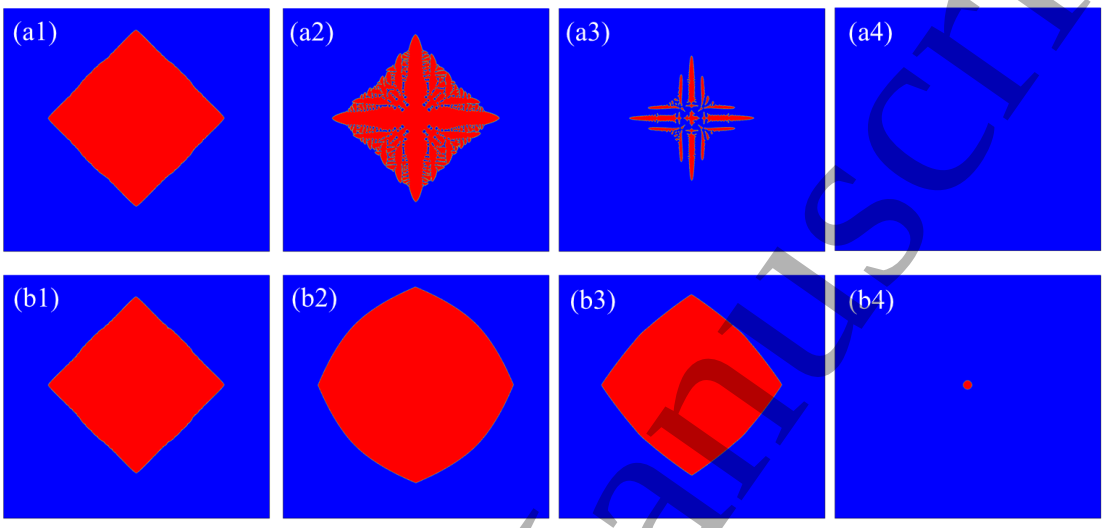


Fig.4 Dendrite morphology at the same time step induced by different current densities applying on Sn and Bi. a1-a4 is the morphology formed in Sn melt with current density of 0, 10 A/m^2, 20 A/m^2, 30 A/m^2, respectively. b1-b4 is the morphology formed in Bi melt with current density of 0, 10 A/m^2, 20 A/m^2, 30 A/m^2, respectively.

3.3 Morphology stability of the solid-liquid interface under electric field

Form discussion in previous studies[8-10], the applied electric field reduces the temperature gradient on the liquid-solid interface, and may destabilize the interface or enhance its instability, so we should firstly analyze the thermoelectric effects on temperature field during solidification. In the case of a pure substance, the solidifying process is governed entirely by the heat flow. That is, the rate of solidification at any point along the liquid-solid interface is governed by how rapidly the heat generated (or absorbed) at that point can be conducted into the bulk of the sample or removed at the boundaries.

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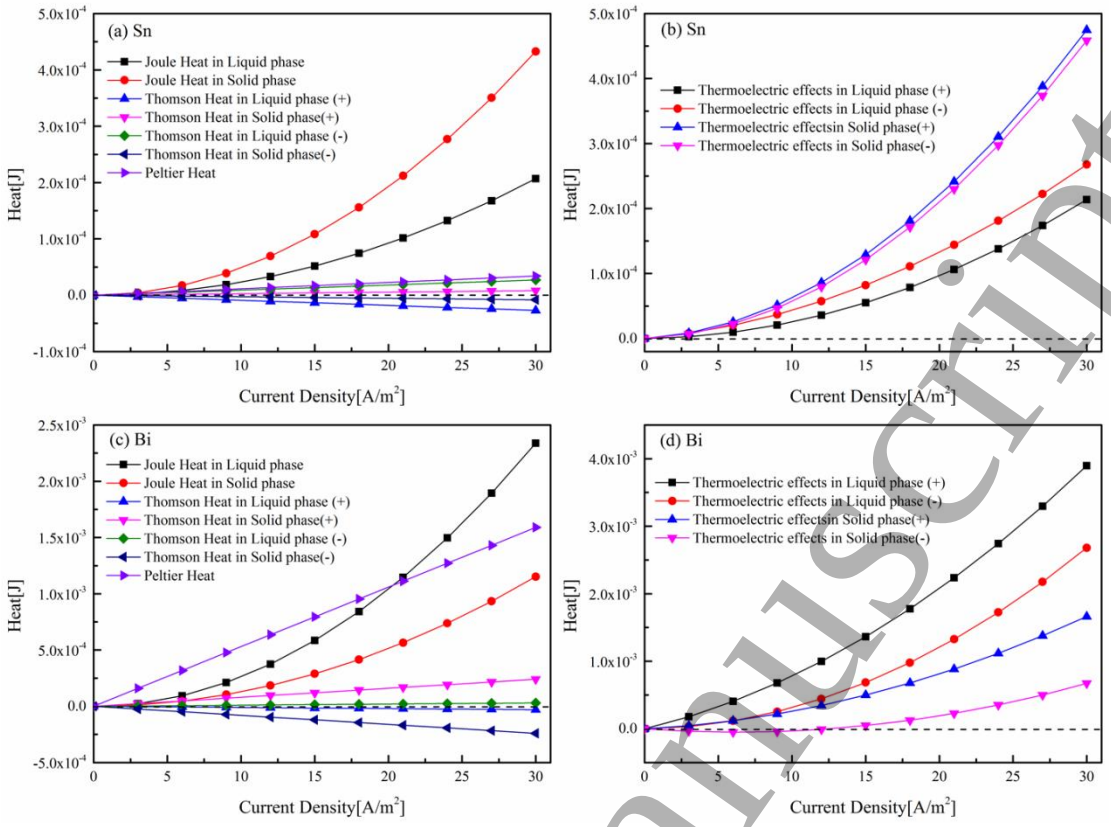


Figure 5 Thermoelectric effects heat induced by different electric current. (a) and (b) is for Sn. (c) and (d) is for Bi.

In Fig.5, we give the calculated heat densities induced by increasing current density for two different metals. From this calculations, we can easily find that: Joule heating is proportional to the square of the electric current density, and serves as a heat source for the system. The Thomson effect is proportional to the vector product of the electric current density and the temperature gradient. Its influence on the energy balance depends on the directions of the electric current density $\boldsymbol{J}$ and the temperature gradient ΔT. The maximum effect is achieved when the two vectors are parallel to each other. For metals with different parameters as shown in Table 2, the heat effect shows different relations with increasing current densities. For Sn, the heat induced with various currents all lead to heat input to the system, and for Bi with negative current densities less than 12, a heat extraction effect takes

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place in the solid phase. And it should be noticed that the magnitude of heat induced by currents for different metals is different, which would lead to different temperature changes in simulation zone. These heat effects could explain the morphology difference in Fig.1 and Fig.4.

4. Concluding remarks

The application of DC has been widely used to control and modify the solidification. Temperature distribution during solidification would be greatly influenced due to the Thermoelectric effects induced by the current. In this study, a phase-field model for simulating the microstructure evolution in pure metals with applied DC is proposed. The thermoelectric heat (including the Joule heat, Thomson heat and Peltier heat) induced by applied DC and latent heat release during liquid-solid transition are all considered in this model. Effect of different magnitude DC on the planar evolution as well as dendrite growth in pure Sn and Bi is simulated, and results indicate that DC has very different influence on the morphology transitions for different metals. It is established that microstructure changes during solidifications is directly attributable to temperature gradient in the growing direction. Thus, the Joule heat, Thomson heat and Peltier heat and latent heat are calculated, and their effects on the temperature distributions near solidifying interfaces is analyzed to explain the effect of different DC on the stability of the solid-liquid interfaces. Therefore, applied DC has a signification effect on the microstructure formations in pure metals by changing the stability of the solid/liquid interfaces with thermoelectric effects.

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